

COMP 3311: Database Management Systems

Lecture 20 Exercises Query Processing: Cost Estimates

Exercise 1: Use the following information to estimate the page I/O cost to compute the query result and the number of query result pages for the query.

Sailor(sailorId: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

**select distinct sName
from Sailor;**

The Sailor relation consists of 60,000 tuples sorted on sailorId. The tuple size is 26 bytes. The page size is 1200 bytes so $bf_{\text{Sailor}} = \lfloor 1200/26 \rfloor = \lfloor 46.15 \rfloor = 46$. Therefore, $B_{\text{Sailor}} = \lceil 60000/46 \rceil = \lceil 1,304.34 \rceil = 1,305$ pages. The buffer size M is 250 pages and 4% of sailor names are the same. There is no index

Sort page I/O cost:

Merge page I/O cost:

Projection page I/O cost:

Query result pages:

Exercise 2: Use the following information about the relations to estimate the page I/O cost to compute the query result using the stated join strategies.

Sailor(sailorId: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

Reserves(sailorId: 2 bytes, boatId: 2 bytes, rDate: 7 bytes)

Query:

```
select *
from Sailor natural join Reserves;
```

Page size: 500 bytes; buffer size M : 100 pages; pointer size: 4 bytes.

n_{Sailor} : 20,000 tuples; $bf_{\text{Sailor}} = \lfloor 500 / 26 \rfloor = 19$ tuples/page; $B_{\text{Sailor}} = \lceil 20,000 / 19 \rceil = 1,053$ pages

n_{Reserves} : 60,000 tuples; $bf_{\text{Reserves}} = \lfloor 500 / 11 \rfloor = 45$ tuples/page; $B_{\text{Reserves}} = \lceil 60,000 / 45 \rceil = 1,334$ pages.

Since there are 20,000 Sailor tuples and 60,000 Reserves tuples, a sailor has, on average, 3 reservations.

a) optimized block nested-loop join

i. using Sailor as the *outer relation*

Query page I/O cost:

ii. using Reserves as the *outer relation*

Query page I/O cost:

b) indexed nested-loop join with a hash index on Reserves.sailorId (*assume no overflow*)

Query page I/O cost:

c) merge join (assume that both relations are not sorted initially)

i. Sort cost

Page I/O cost to sort Sailor:

Page I/O cost to sort Reserves:

Total page I/O cost to sort the relations:

ii. Merge cost (to join the relations)

Total page I/O cost to join the relations:

Query page I/O cost:

Exercise 3: Use the following information about the relations to process the query.

Sailor(sailorId: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

Reserves(sailorId: 2 bytes, boatId: 2 bytes, rDate: 7 bytes)

The Sailor relation contains 40,000 tuples in 262 pages and the Reserves relation contains 100,000 tuples in 276 pages. There are 10 different sailor ratings and 100 different boats. Assume that sailors are distributed uniformly over the 10 ratings and reservations are distributed uniformly over the 100 boats. The page size is 4000 bytes.

The Sailor relation contains a clustering B⁺-tree index with 3 levels on sailorId, and the Reserves relation contains a hash index on boatId. A page can fit 400 record pointers. For non-clustering indexes, each record pointer is assumed to point to a different page. We want to process the query given below.

Query:

```
select sName
from Sailor natural join Reserves
where boatId=30
and rating>5;
```

Some useful statistics

- On average, each sailor has 2.5 reservations.
- On average, each boat has 1,000 reservations.
- On average, for each rating there are 4,000 sailors.

Estimate the minimum page I/O cost of processing the query using a fully pipelined execution method (i.e., do not materialize anything except the query result).

Base Strategy:

Page I/O cost:

a) Cost to evaluate $\sigma_{\text{boatId}=30}$ Reserves

Strategy a1: linear search on Reserves

Page I/O cost:

Strategy a2: index lookup using hash index on Reserves.boatId

Page I/O cost:

b) Cost to evaluate $\sigma_{\text{rating}>5}$ Sailor (no index on rating)

Strategy b1: linear search on Sailor

Page I/O cost:

Strategy b2: check the condition in the buffer (on-the-fly) during the join

Page I/O cost:

c) Cost to evaluate Sailor $\bowtie$ Reserves

Strategy c1: block nested-loop (following Strategy b1)

Page I/O cost:

Strategy c2: block nested-loop (following Strategy b2)

Page I/O cost:

Strategy c3: indexed nested-loop using sailorId index on Sailor (following Strategy b2)

Page I/O cost:

d) Cost to evaluate π_{sName}

Strategy d1: perform the projection in the buffer (on-the-fly) during the join

Page I/O cost:

Query page I/O cost:

Name: (1) _____ / _____ Student#: (1) _____ Date: _____
Family/Given (PRINT) Given/First (PRINT)

Name: (2) _____ / _____ Student#: (2) _____
Family/Given (PRINT) Given/First (PRINT)

These are partner exercises. YOU **MUST DO THESE EXERCISES WITH A PARTNER**. Individual submissions will not be graded.
A genuine effort must be made to complete all exercises to obtain full marks.
SUBMIT ONLY ONE COMPLETED PARTNER WORKSHEET.

COMP 3311: Database Management Systems

Lecture 20 Exercises Query Processing: Cost Estimates

Exercise 4: Use the following information to estimate the page I/O cost to compute the query result and the number of query result pages for the query.

**select distinct sName
from Sailor;**

Sailor(sailorId: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

The Sailor relation consists of 40,000 tuples sorted on sailorId. The tuple size is 26 bytes. The page size is 800 bytes so $bf_{Sailor} = \lceil 800/26 \rceil = 30$. Therefore, $B_{Sailor} = \lceil 40,000/30 \rceil = 1,334$ pages. The buffer size M is 100 pages and 5% of sailor names are the same. There is no index.

Sort page I/O cost:

Merge page I/O cost:

Projection page I/O cost:

Query result pages:

Name: (1) _____ / _____ Student#: (1) _____ Date: _____
Family/Given (PRINT) Given/First (PRINT)

Name: (2) _____ / _____ Student#: (2) _____
Family/Given (PRINT) Given/First (PRINT)

These are partner exercises. YOU **MUST DO THESE EXERCISES WITH A PARTNER**. Individual submissions will not be graded.
A genuine effort must be made to complete all exercises to obtain full marks.
SUBMIT ONLY ONE COMPLETED PARTNER WORKSHEET.

COMP 3311: Database Management Systems

Lecture 20 Exercises Query Processing: Cost Estimates

Exercise 5: Use the following information about the relations to estimate the page I/O cost to compute the query result using the stated join strategies.

Sailor(sailorId: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

Reserves(sailorId: 2 bytes, boatId: 2 bytes, rDate: 7 bytes)

Page size: 1,000 bytes; buffer size M : 100 pages; pointer size: 4 bytes

n_{Sailor} : 10,000 tuples; n_{Reserves} : 40,000 tuples

Query:

```
select *  
from Sailor natural join Reserves;
```

a) query page I/O cost for optimized block nested-loop join

i. using Sailor as the *outer relation*

ii. using Reserves as the *outer relation*

b) query page I/O cost for indexed nested-loop join with a hash index on Reserves.sailorId (*no overflow*)

c) query page I/O cost for merge join (assume that both relations are not sorted initially)

i. Sort cost

Page I/O cost to sort Sailor:

Page I/O cost to sort Reserves:

Total page I/O cost to sort the relations:

ii. Merge cost (to join the relations)

Total page I/O cost to join the relations:

iii. Query page I/O cost:

Name: (1) _____ / _____ Student#: (1) _____ Date: _____
Family/Given (PRINT) Given/First (PRINT)

Name: (2) _____ / _____ Student#: (2) _____
Family/Given (PRINT) Given/First (PRINT)

These are **partner** exercises. **YOU MUST DO THESE EXERCISES WITH A PARTNER.** Individual submissions will not be graded.
A genuine effort must be made to complete all exercises to obtain full marks.
SUBMIT ONLY ONE COMPLETED PARTNER WORKSHEET.

Exercise 6: Use the following information about the relations to estimate the page I/O cost to compute the query result.

Student(studentId: 8 bytes, firstName: 20 bytes, lastName: 25 bytes, email: 15 bytes, unitId: 4 bytes)

Enrollsn(studentId: 8 bytes, courseId: 8 bytes, grade: 4 bytes)

The Student relation contains 10,000 tuples in 1,000 pages and the Enrollsn relation contains 50,000 tuples in 1,352 pages. The page size is 1,000. There are 25 different academic units and 1,000 different courses.

The Student relation contains a clustering index on unitId and the Enrollsn relation contains a non-clustering index on studentId. Each available index is a B⁺-tree with 3 levels. For non-clustering indexes, each record pointer leads to a different page.

Query:

```
select firstName, lastName
from Student natural join Enrollsn
where courseId='COMP 3311'
and unitId='COMP';
```

Some useful statistics

- On average, a student enrolls in 5 courses.
- On average, an academic unit has 400 students.
- On average, each course has an enrollment of 50 students.

Estimate the minimum page I/O cost of processing the query using a fully pipelined execution method (i.e., do not materialize anything except the query result).

Base Strategy:

Page I/O cost:

a) Cost to evaluate $\sigma_{\text{courseId}='COMP 3311'}$ Enrollsn (no index on courseId)

Best strategy:

Page I/O cost:

b) Cost to evaluate $\sigma_{\text{unitId}='COMP'}$ Student

Best strategy:

Page I/O cost:

c) Cost to evaluate Student $\bowtie$ Enrollsn

Best strategy:

Page I/O cost:

d) Cost to evaluate $\pi_{\text{firstName, lastName}}$

Best strategy:

Page I/O cost:

Query page I/O cost: